

Vulnerability Assessment Report

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Task: Task 1

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1. Introduction

A vulnerability assessment is the process of identifying, analyzing, and reporting security weaknesses in a system or website. This report evaluates a sample web application to identify potential vulnerabilities and provide recommendations to improve security.

2. Target Information

- Target Website: <http://testphp.vulnweb.com>
- Type: Test Web Application
- Purpose: Security testing and learning

3. Tools Used

- Nmap (Network scanning)
- OWASP ZAP (Passive vulnerability scanning)
- Browser Developer Tools

4. Methodology

The following steps were followed during the assessment:

1. Scanned the target using Nmap to identify open ports and services
2. Performed passive scanning using OWASP ZAP
3. Manually inspected the website using browser developer tools
4. Identified potential vulnerabilities
5. Classified risks based on severity

5. Findings

5.1 Open Ports Detected

- Port 80 (HTTP) is open

Risk Level: Low

Description: Open ports can expose services to attackers

Recommendation:

Restrict unnecessary open ports and use firewall rules

5.2 Missing Security Headers

- X-Frame-Options not set
- Content-Security-Policy missing

Risk Level: Medium

Description: Missing headers can make the website vulnerable to attacks like clickjacking

Recommendation:

Implement proper security headers in server configuration

5.3 Cross-Site Scripting (XSS) Possibility

- User input fields are not properly sanitized

Risk Level: High

Description: Attackers can inject malicious scripts into the website

Recommendation:

Validate and sanitize all user inputs

Use output encoding

5.4 Information Disclosure

- Server details are visible

Risk Level: Low

Description: Revealing server information can help attackers

Recommendation:

Hide server version details

6. Risk Summary

Vulnerability	Risk Level
Open Ports	Low
Missing Security Headers	Medium
XSS	High
Information Disclosure	Low

7. Conclusion

The vulnerability assessment identified several security issues, including missing security headers and potential XSS vulnerabilities. These issues can be exploited by attackers if not addressed. Implementing proper security measures will significantly improve the overall security of the web application.

8. References

- OWASP Top 10
 - Nmap Documentation
 - OWASP ZAP Tool
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